

The Structural Polynomial of a Parity Sequence and Its Applications

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Abstract

We develop a polynomial representation for parity sequences of generalised Collatz-type maps (g, h, q) , working in the ring $\mathbb{Z}[g, h]$ rather than specialising immediately to numerical values. The central object is the *structural polynomial* $k_w(g, h)$ attached to a parity sequence, which encodes the accumulated additive contribution of the odd steps independently of the parameter q .

The main results are: (1) a *path identity* $h^e b = g^o a + q k_w(g, h)$ valid for any parity sequence; (2) a *geometric polynomial theorem* showing that Steiner words $\text{St}(\alpha, \beta)$ have $k(g, h) = (g^\alpha - h^\alpha)/(g - h)$, independent of β ; (3) a *composition rule* $k_{Q \circ P} = g^{o_Q} k_P + h^{e_P} k_Q$; and (4) a *cycle element identity* $a = q k_w/(h^e - g^o)$ giving the fixed point as a rational function of (g, h, q) .

Beyond these core results, we develop a system of five equivalent representations of any parity sequence — its bits (the p -value), the structural polynomial $k_w(g, h)$, its specialisation $k_w(3, 2)$, the σ -polynomial $\sigma_w(u, v)$, and the orbit parity sequence ϕ_w — connected by explicitly invertible transformations. We also introduce a successor operation on k_w in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$ that generates all cycle representatives by rotation, and we warn that polynomial non-divisibility of $h^e - g^o \nmid k_w$ in $\mathbb{Z}[g, h]$ does not preclude pointwise divisibility at specific integer bases. Applications to the $2a + 1$ composition identity are developed in the companion paper [4].

Contents

1	Introduction	2
2	The Structural Polynomial and Path Identity	3
3	Steiner Words and the Geometric Polynomial Theorem	4
4	Composition of Structural Polynomials	5

5	Affine Path Equations	6
5.1	The admissibility constraint	6
5.2	The affine path family	6
5.3	Intersection with cycles	7
6	The Cycle Element Identity	8
7	Cycle Flavours	12
8	Representations of a Parity Sequence	13
8.1	The p -value (edges B, C; centre node)	14
8.2	The σ -polynomial (edges A, D)	15
8.3	Worked example (all edges, $w = OEEOEOEE$)	16
9	What the Abstract Cycle Tells Us — and What It Does Not	17
10	Examples	18
11	Related Work	20

1 Introduction

A *generalised Collatz-type map* with basis (g, h, q) consists of two elementary operations on the positive integers: an *odd step* $x \mapsto gx + q$ and an *even step* $x \mapsto x/h$. The standard Collatz map (the Syracuse odd-only iteration) uses basis $(g, h, q) = (3, 2, 1)$; the $5x + 1$ analogue uses $(5, 2, 1)$. A *parity sequence* is a finite word over $\{O, E\}$ specifying the order in which these steps are applied. Each parity sequence acts as an affine map on its domain of definition.

The central algebraic object we study is the *structural polynomial* $k(g, h) \in \mathbb{Z}[g, h]$ attached to a parity sequence. It captures the additive part of the parity sequence’s affine action in a basis-independent form. The key advantage of working in $\mathbb{Z}[g, h]$ rather than substituting $g = 3$, $h = 2$ at the outset is that polynomial identities can be proved and, just as importantly, their *failure for other bases* can be identified precisely. A numerical check that an identity holds at $(3, 2)$ gives no information about why it fails at $(5, 2)$; working symbolically makes the obstruction explicit.

The power of the symbolic approach is illustrated dramatically by the $5x + 1$ system, which possesses a non-trivial cycle $17 \rightarrow 43 \rightarrow 27 \rightarrow 17$ under the Syracuse $5x + 1$ map. Via the cycle element identity (Section 6), the fixed point $a = 17$ emerges as a rational function of (g, h, q) : it equals $q \cdot k(g, h)/(h^e - g^o)$, where $k(5, 2) = 51$ and $h^7 - g^3 = 128 - 125 = 3$, giving $a = 51/3 = 17$. The near-miss $h^e - g^o = 3$ is the arithmetic miracle that makes the cycle possible, and it is visible as a polynomial phenomenon.

The paper is organised as follows. Section 2 establishes the path identity and defines the structural polynomial. Section 3 proves the geometric polynomial theorem for Steiner words. Section 4 derives the composition rule. Section 6 applies these tools to cycle element identification, and includes a remark cautioning that polynomial non-divisibility does

not imply pointwise non-divisibility. Section 8 develops the five representations of a parity sequence — the p -value, the σ -polynomial, $k_w(g, h)$, $k_w(3, 2)$, and the orbit parity sequence ϕ_w — and unifies them in a single diagram. Section 10 collects worked examples illustrating the theory.

2 The Structural Polynomial and Path Identity

Definition 2.1 (Basis and elementary maps). A *basis* is a triple (g, h, q) of integers with $g, h \geq 1$ and $q \neq 0$. The associated elementary maps are the *odd step* $O: x \mapsto gx + q$ and the *even step* $E: x \mapsto x/h$.

Definition 2.2 (Parity sequence). A *parity sequence* is a finite word $w = w_1 w_2 \cdots w_n$ (possibly empty, $n \geq 0$) over the alphabet $\{O, E\}$. It defines an affine map Φ_w by composing the corresponding elementary maps from left to right: $\Phi_w = \Phi_{w_n} \circ \cdots \circ \Phi_{w_1}$ (the empty composition is the identity map). We write $o = o(w)$ for the number of occurrences of O in w and $e = e(w)$ for the number of occurrences of E .

Remark 2.3. Throughout this paper we assume $h \mid x$ whenever an E -step is applied, so that all maps are defined on their natural domain.

Definition 2.4 (κ -tuple). Let w be a parity sequence with $o \geq 1$ odd steps and e even steps. Number the O -steps from 0 to $o - 1$ in the order they appear in w . For $0 \leq j \leq o - 1$, let κ_j denote the number of E -steps that appear in w *strictly before* the $(j + 1)$ -th O -step. The sequence $(\kappa_0, \kappa_1, \dots, \kappa_{o-1})$ is the κ -tuple of w . It satisfies $0 \leq \kappa_0 \leq \kappa_1 \leq \cdots \leq \kappa_{o-1} \leq e$. For example, $w = OEEOEEOEE$ has $o = 3$ odd steps at positions 0, 3, 5, preceded by 0, 2, 3 even steps respectively, so $\kappa = (0, 2, 3)$.

Remark 2.5. For the standard Collatz basis $(g = 3, q = 1)$, a parity sequence beginning with an O -step (i.e. $\kappa_0 = 0$) corresponds to a legal orbit starting from an odd integer if and only if its κ -tuple is strictly increasing (equivalently, no two O -steps are adjacent). The polynomial framework works unchanged for non-strict profiles (see Example 10.3).

Definition 2.6 (Structural polynomial). The *structural polynomial* of a parity sequence w with κ -tuple $(\kappa_0, \dots, \kappa_{o-1})$ is

$$k_w(g, h) = \sum_{j=0}^{o-1} g^{o-1-j} h^{\kappa_j} \in \mathbb{Z}[g, h].$$

When $o = 0$, we set $k_w = 0$.

We can now state and prove the central algebraic identity.

Theorem 2.7 (Path identity). *Let (g, h, q) be a basis and w a parity sequence with o odd steps and e even steps, having structural polynomial $k_w(g, h)$. If $b = \Phi_w(a)$ then*

$$h^e \cdot b = g^o \cdot a + q \cdot k_w(g, h). \tag{1}$$

Proof. We proceed by induction on the length $n = |w|$ of the parity sequence.

Base cases. If $w = O$ (single odd step), then $e = 0$, $o = 1$, $b = ga + q$, and $k_w = g^0 h^0 = 1$. The identity reads $b = ga + q \cdot 1$, which holds. If $w = E$ (single even step), then $o = 0$, $e = 1$, $b = a/h$, $k_w = 0$, and the identity reads $h \cdot (a/h) = a$, which holds.

Inductive step. Suppose (1) holds for all parity sequences of length less than n , and write $w = w' \cdot \sigma$ where $\sigma \in \{O, E\}$ is the last step. Let $c = \Phi_{w'}(a)$ be the intermediate value, so $b = \Phi_\sigma(c)$. By the inductive hypothesis,

$$h^{e'} \cdot c = g^{o'} \cdot a + q \cdot k_{w'}(g, h), \quad (2)$$

where $o' = o(w')$ and $e' = e(w')$.

Case $\sigma = E$. Then $b = c/h$, $o = o'$, and $e = e' + 1$. Since $h^{e'+1}(c/h) = h^{e'}c = g^{o'}a + qk_{w'}$ by (2), and the E-step appended at the end does not precede any O-step, so the κ -tuple of w equals that of w' , and therefore $k_w = k_{w'}$. The identity (1) holds.

Case $\sigma = O$. Then $b = gc + q$, so $o = o' + 1$ and $e = e'$. The new O-step is the last (index $j = o'$) in w . All $e' = e$ even steps in w precede it (they all belong to w'), so $\kappa_{o'} = e'$. The structural polynomial of w is

$$k_w(g, h) = g \cdot k_{w'}(g, h) + h^{e'},$$

because the exponents of the first o' terms each gain one extra factor of g , and the new final O-step contributes $g^0 h^{e'} = h^{e'}$. Now compute:

$$\begin{aligned} h^e \cdot b &= h^{e'} \cdot (gc + q) \\ &= g \cdot h^{e'} c + q \cdot h^{e'} \\ &= g(g^{o'} a + q \cdot k_{w'}(g, h)) + q \cdot h^{e'} \quad (\text{by (2)}) \\ &= g^{o'+1} a + q(g \cdot k_{w'}(g, h) + h^{e'}) \\ &= g^o \cdot a + q \cdot k_w(g, h). \end{aligned} \quad \square$$

Remark 2.8. Two features of the path identity deserve emphasis.

1. The polynomial $k_w(g, h)$ is independent of q : the parameter q appears only as a uniform scalar multiplier. Thus the additive structure of parity sequences is entirely captured by the polynomial ring $\mathbb{Z}[g, h]$.
2. The polynomial $k_w(g, h)$ is independent of trailing E -steps: appending E -steps at the end only scales h^e on the left-hand side of (1), leaving k_w unchanged. This means k_w alone does not uniquely identify a parity sequence; the p -value (Section 8) resolves the ambiguity by appending a sentinel bit that records the length n .

3 Steiner Words and the Geometric Polynomial Theorem

Definition 3.1 (Steiner word). For integers $\alpha \geq 1$ and $\beta \geq 0$, the *Steiner word* $\text{St}(\alpha, \beta)$ is the parity sequence

$$\text{St}(\alpha, \beta) = \underbrace{(OE)^\alpha}_{\alpha \text{ copies of } OE} \underbrace{E^\beta}_{\beta \text{ extra } E\text{-steps}}.$$

It has $o = \alpha$ odd steps and $e = \alpha + \beta$ even steps.

Remark 3.2. Steiner words are the natural building blocks of Collatz orbits under the Syracuse (odd-only) map: each Syracuse step $S(a) = (3a + 1)/2^{v_2(3a+1)}$ corresponds to a Steiner word $\text{St}(1, \beta)$ where $\beta = v_2(3a + 1) - 1 \geq 0$. A block of α consecutive applications of S before the next element divisible by an extra power of h corresponds to $\text{St}(\alpha, \beta)$.

The κ -tuple of $\text{St}(\alpha, \beta)$ has a particularly simple form.

Lemma 3.3 (κ -tuple of a Steiner word). *The κ -tuple of $\text{St}(\alpha, \beta)$ is $\kappa_j = j$ for $0 \leq j \leq \alpha - 1$.*

Proof. In $(OE)^\alpha E^\beta$, the j -th O -step (zero-indexed) appears at position $2j + 1$. Exactly j E -steps appear before it (one after each of the first j O -steps). Hence $\kappa_j = j$. $\square$

Theorem 3.4 (Geometric polynomial theorem). *For any $\alpha \geq 1$ and $\beta \geq 0$,*

$$k_{\text{St}(\alpha, \beta)}(g, h) = \frac{g^\alpha - h^\alpha}{g - h} = \sum_{j=0}^{\alpha-1} g^{\alpha-1-j} h^j. \quad (3)$$

In particular, $k_{\text{St}(\alpha, \beta)}$ is independent of β .

Proof. Substituting $\kappa_j = j$ from Lemma 3.3 into Definition 2.6:

$$k_{\text{St}(\alpha, \beta)}(g, h) = \sum_{j=0}^{\alpha-1} g^{\alpha-1-j} h^j = \frac{g^\alpha - h^\alpha}{g - h},$$

where the last equality is the standard geometric series identity. Independence of β follows from Remark 2.8. $\square$

4 Composition of Structural Polynomials

If P and Q are parity sequences, the composition $Q \circ P$ denotes the parity sequence obtained by concatenating P then Q : first apply P , then apply Q to the result.

Theorem 4.1 (Composition rule). *Let P and Q be parity sequences with o_P, e_P and o_Q, e_Q odd and even steps respectively. Then*

$$k_{Q \circ P}(g, h) = g^{o_Q} \cdot k_P(g, h) + h^{e_P} \cdot k_Q(g, h). \quad (4)$$

Proof. Let $(\kappa_0^P, \dots, \kappa_{o_P-1}^P)$ and $(\kappa_0^Q, \dots, \kappa_{o_Q-1}^Q)$ be the κ -tuples of P and Q respectively.

In the concatenated sequence $Q \circ P$, the first o_P odd steps come from P and the last o_Q odd steps come from Q . The E -steps of P all precede the E -steps of Q .

For the j -th odd step of P ($0 \leq j < o_P$): the number of E -steps preceding it in $Q \circ P$ equals κ_j^P (no E -steps from Q are involved).

For the j -th odd step of Q ($0 \leq j < o_Q$): the number of E -steps preceding it in $Q \circ P$ is $e_P + \kappa_j^Q$ (all e_P steps of P plus the κ_j^Q steps of Q that precede it).

Therefore the structural polynomial of $Q \circ P$ is

$$\begin{aligned}
k_{Q \circ P}(g, h) &= \sum_{j=0}^{o_P-1} g^{(o_P+o_Q-1)-j} h^{\kappa_j^P} + \sum_{j=0}^{o_Q-1} g^{o_Q-1-j} h^{e_P+\kappa_j^Q} \\
&= g^{o_Q} \sum_{j=0}^{o_P-1} g^{o_P-1-j} h^{\kappa_j^P} + h^{e_P} \sum_{j=0}^{o_Q-1} g^{o_Q-1-j} h^{\kappa_j^Q} \\
&= g^{o_Q} \cdot k_P(g, h) + h^{e_P} \cdot k_Q(g, h). \quad \square
\end{aligned}$$

Remark 4.2 (Interpretation). The two terms in (4) have natural interpretations. The factor g^{o_Q} on k_P accounts for the fact that after P has run, Q applies o_Q further odd multiplications, each of which scales the additive accumulation from P by g . The factor h^{e_P} on k_Q accounts for the fact that Q 's own E -accumulation exponents are displaced upward by the e_P even divisions already consumed by P .

5 Affine Path Equations

The path identity (1) can be read in two directions. Read as a statement about a specific pair (a, b) , it asserts that starting from a and following word w arrives at b . Read as a constraint, it characterises the full family of integer pairs (a, b) for which w is a valid path.

5.1 The admissibility constraint

For a word w with o odd steps and e even steps, the path identity is

$$h^e \cdot b = g^o \cdot a + q \cdot k_w(g, h).$$

For b to be a positive integer we need $h^e \mid g^o a + q k_w(g, h)$. Since $\gcd(g^o, h^e) = 1$ when $\gcd(g, h) = 1$, this is a congruence condition on a alone:

$$g^o \cdot a \equiv -q k_w(g, h) \pmod{h^e}.$$

Because $\gcd(g, h) = 1$ implies g^o is invertible modulo h^e , this has a unique solution $r \in \{0, 1, \dots, h^e - 1\}$:

$$r \equiv -q k_w(g, h) \cdot (g^o)^{-1} \pmod{h^e}.$$

We call r the *admissibility residue* of w .

5.2 The affine path family

Every valid starting point a for a path with word w satisfies $a \equiv r \pmod{h^e}$, so the family of integer pairs is parameterised by $t \geq 0$:

$$a(t) = h^e \cdot t + r, \quad b(t) = g^o \cdot t + b_0, \quad (5)$$

where $b_0 = (g^o r + q k_w(g, h))/h^e$ is the image of r under Φ_w . Both $a(t)$ and $b(t)$ are affine functions of t ; the path $a \rightarrow b$ is the letter-driven trajectory prescribed by w .

Remark 5.1. The residue r encodes the admissibility of the path: a starting value a is compatible with word w if and only if $a \equiv r \pmod{h^e}$. This is the modular shadow of the requirement that each prescribed E-step is applied to an even value.

Example 5.2 (St(2, 1) at the Collatz basis). For $w = \text{St}(2, 1) = \text{OE0EE}$ at $(g, h, q) = (3, 2, 1)$: $o = 2$, $e = 3$, $k_w(3, 2) = g + h = 5$, $h^e = 8$, $g^o = 9$. Since $9 \equiv 1 \pmod{8}$, the admissibility residue is $r \equiv -5 \cdot 1 \equiv 3 \pmod{8}$. The affine family is $a(t) = 8t + 3$, $b(t) = 9t + 4$, giving the paths $3 \rightarrow 4$, $11 \rightarrow 13$, $19 \rightarrow 22$, $27 \rightarrow 31$, $\dots$

5.3 Intersection with cycles

Setting $b = a$ in (5) requires

$$h^e \cdot t + r = g^o \cdot t + b_0,$$

hence $(h^e - g^o) \cdot t = b_0 - r$, i.e. $d \cdot t = b_0 - r$. This has an integer solution in t if and only if $d \mid (b_0 - r)$, and when it does, the solution is unique:

$$t^* = \frac{b_0 - r}{d}, \quad a^* = h^e \cdot t^* + r.$$

Substituting $t^* = (b_0 - r)/d$ into $a^* = h^e t^* + r$ and using $b_0 = (g^o r + qk)/h^e$:

$$d \cdot a^* = h^e b_0 - g^o r = (g^o r + qk) - g^o r = qk.$$

The residue r cancels exactly, recovering the cycle element identity (6). The affine path family thus makes explicit what the cycle identity summarises: a cycle exists when the path family and the diagonal $a = b$ intersect at a non-negative integer t^* , and at that intersection the admissibility residue drops out.

Remark 5.3. For path problems, t ranges freely over $\mathbb{Z}_{\geq 0}$, giving an infinite arithmetic progression of valid (a, b) pairs. For cycle problems, t is pinned to the single value t^* (if it exists and is a non-negative integer). The transition from path to cycle is therefore a transition from a free parameter to a determined one — the same phenomenon seen in the cycle element identity, here made geometric.

Example 5.4 (The $5x + 1$ cycle at $a = 17$). Consider $w = \text{OE0EEEE0EEE}$ at basis $(g, h, q) = (5, 2, 1)$: $o = 3$, $e = 7$, $h^e = 128$, $g^o = 125$, $d = 3$, $k_w(5, 2) = 51$.

The admissibility residue is $r \equiv -51 \cdot (125)^{-1} \pmod{128}$. Since $125 \equiv -3 \pmod{128}$ and $(-3)^{-1} \equiv 85 \pmod{128}$ (as $(-3)(85) = -255 \equiv 1 \pmod{128}$):

$$r \equiv -51 \cdot 85 = -4335 \equiv 17 \pmod{128}.$$

The image of r under Φ_w is $b_0 = (125 \cdot 17 + 51)/128 = 17$, so the affine family is

$$a(t) = 128t + 17, \quad b(t) = 125t + 17.$$

Setting $a = b$: $3t = b_0 - r = 0$, giving $t^* = 0$ and $a^* = 17$. The equation $3t = 0$ has exactly one non-negative integer solution, so $a = 17$ is the unique element of this cycle.

6 The Cycle Element Identity

Setting $a = b$ in the path identity (1) immediately gives the *cycle element identity*: for any fixed point $\Phi_w(a) = a$,

$$(h^e - g^o) \cdot a = q \cdot k_w(g, h). \quad (6)$$

Writing $d = h^e - g^o$ and $k = k_w(g, h)$, the *natural solution* $(a, q) = (k, d)$ is always an integer solution, so every abstract cycle word is immediately realisable. The interesting question is which cycles admit a *reduced* solution — one where $\gcd(a, q) = 1$ and in particular $q = 1$.

Throughout this section $\gcd$ denotes the positive greatest common divisor, so $\gcd(k, |d|) > 0$ always, and the sign of $q_w = d/\gcd(k, |d|)$ matches the sign of d .

Proposition 6.1 (Every abstract cycle has a natural encoding). *Let w be any parity sequence with $o \geq 1$ odd steps and e even steps, and let (g, h) be any integer pair. Setting $q = d = h^e - g^o$ and $a = k = k_w(g, h)$, the pair (a, q) satisfies $\Phi_w(a) = a$ at basis (g, h, q) . In particular, every abstract cycle word w is realised as a genuine integer cycle at every integer pair (g, h) simply by choosing $q = h^e - g^o$.*

Proof. The pair $(a, q) = (k, d)$ satisfies $d \cdot k = d \cdot k$ trivially, hence $d \cdot a = q \cdot k$. Setting $b = a$ in the path identity (1) shows this is equivalent to $\Phi_w(a) = a$. $\square$

Remark 6.2. Since the natural solution exists for every (g, h) and every word w , the existence of an integer cycle is not in itself remarkable — it is guaranteed by construction once $q = d$. What matters is *reduction*: when $\gcd(g, h) = 1$ the factor $f = \gcd(k, |d|)$ is uniform across the whole cycle (Lemma 6.3 below), and dividing through gives the reduced solution $a_w = k/f$, $q_w = d/f$ with $\gcd(a_w, |q_w|) = 1$. The mythical cycles are the reduced cycles at the standard Collatz basis $(g, h, q) = (3, 2, 1)$ with $o > 1$: these would require $|d| = |2^e - 3^o|$ to divide $k_w(3, 2)$, and no such cycle is known to exist.

Lemma 6.3 (Natural cycles carry a uniform factor). *Let $\gcd(g, h) = 1$, let w be a parity sequence with $o \geq 1$, and let $w_0, w_1, \dots, w_{n-1}$ be the cyclic rotations of w with structural values $k_i = k_{w_i}(g, h)$. Set $d = h^e - g^o$ and $f = \gcd(k_0, |d|)$. Then $f \mid k_i$ for every i , and consequently $f \mid \gcd(k_0, k_1, \dots, k_{n-1}, |d|)$. In particular, the reduced solution $a_w = k_0/f$, $q_w = d/f$ is a valid integer cycle.*

Proof. We show $f \mid k_i$ is preserved under each rotation step. By hypothesis $f \mid k_0$ and $f \mid |d|$.

O-leading step (w_i begins with O): the successor value satisfies $k_{i+1} = gk_i + d$ (Proposition 6.10). Since $f \mid k_i$ and $f \mid d$, we have $f \mid gk_i + d = k_{i+1}$.

E-leading step (w_i begins with E): the successor value satisfies $k_{i+1} = k_i/h$. Since $\gcd(g, h) = 1$, we have $\gcd(h, d) = \gcd(h, h^e - g^o) = \gcd(h, g^o) = 1$, so $\gcd(f, h) = 1$. Together with $f \mid k_i$ and $h \mid k_i$ (required for the E-step to yield an integer), $\gcd(f, h) = 1$ gives $f \mid k_i/h = k_{i+1}$.

Since f divides every k_i and $|d|$, the reduced pair $(k_0/f, d/f)$ satisfies $d \cdot (k_0/f) = (d/f) \cdot k_0$, confirming it is a valid integer cycle. $\square$

Remark 6.4. This lemma explains the role of $\gcd(g, h) = 1$ in the reduction argument. It is not that reduced cycles fail to exist when $\gcd(g, h) > 1$ — they may still exist — but the propagation argument above breaks down at the E-leading step: $\gcd(f, h) = 1$ need not hold

when h and d share a common factor, so $f \mid k_i$ does not automatically imply $f \mid k_i/h$. When $\gcd(g, h) = 1$, every natural cycle carries the uniform factor $f = \gcd(k_0, |d|)$, and dividing through always produces a reduced cycle.

Lemma 6.5 (Cycle gcd under non-coprime bases). *Let $w = w_0 w_1 \cdots w_{n-1}$ be a parity sequence with o odd steps, e even steps, $d = h^e - g^o$, and let $r = \gcd(g, h)$. If $r > 1$ then $r \mid k_{w_i}(g, h)$ for every rotation w_i of w , and $r \mid d$. Consequently $\gcd(k_{w_i}(g, h), d)$ is divisible by r for each i , and in general*

$$\gcd(k_{w_i}(g, h), d) \neq \gcd(k_{w_j}(g, h), d)$$

for distinct rotations $i \neq j$. The primitive solution to the cycle system therefore requires the joint gcd

$$\delta_w := \gcd(k_{w_0}(g, h), k_{w_1}(g, h), \dots, k_{w_{n-1}}(g, h), |d|),$$

not the single-rotation gcd $\gcd(k_{w_i}(g, h), d)$. When $\gcd(g, h) = 1$, the single-rotation gcd suffices: $\delta_w = \gcd(k_w(g, h), |d|)$ for any rotation w .

Proof. Every monomial $g^{o-1-j} h^{\kappa_j}$ of $k_w(g, h)$ has positive total degree in g and h (since $o \geq 1$), so $r \mid k_{w_i}$ for every rotation w_i . Since $r \mid g^o$ and $r \mid h^e$, we have $r \mid d$. Hence r divides every individual $\gcd(k_{w_i}, d)$, and $\gcd(k_{w_i}, d)$ may differ from $\gcd(k_{w_j}, d)$ for $i \neq j$ because the spurious factor r is distributed unevenly across rotation values. The joint gcd δ_w absorbs only the common content of all rotations and is therefore the correct divisor. When $\gcd(g, h) = 1$, the rotation recurrence (Proposition 6.10) acts by multiplication by g or h^{-1} modulo d , so all k_{w_i} lie in the same $\gcd(\cdot, d)$ -class, and the single-rotation formula suffices. $\square$

Remark 6.6. The failure of the single-rotation formula when $\gcd(g, h) > 1$ is illustrated concretely by $w = \text{OEEOEOEE}$ at $(g, h) = (4, 2)$, where $r = 2$. Here $d = 2^5 - 4^3 = -32$, and the eight structural values $k_{w_i}(4, 2)$ are 32, 96, 48, 160, 80, 40, 128, 64.

Per-rotation reduction (using $\gcd(k_i, d)$ for each i separately) gives inconsistent results:

Rotation	k_i	$\gcd(k_i, d)$	$a_i = k_i / \gcd$	$q_i = d / \gcd$
OEEOEOEE	32	32	1	-1
EOEOEOEO	96	32	3	-1
OEOEOEOE	48	16	3	-2
EEEOEOEO	160	32	5	-1
EOEOEOEO	80	16	5	-2
OEOEOEOE	40	8	5	-4
EEEOEOEO	128	32	4	-1
EOEOEOEO	64	32	2	-1

The q_i values are inconsistent ($-1, -2, -4$) and the a_i values repeat (the value 5 appears three times, 3 twice), so per-rotation reduction does not yield a valid 8-cycle with distinct elements.

Joint-gcd reduction (using $\delta_w = \gcd(32, 96, 48, 160, 80, 40, 128, 64, |d|) = 8$ for all rotations) gives consistent results:

Rotation	k_i	δ_w	$a_i = k_i/\delta_w$	$q_i = d/\delta_w$
OEOEEOE	32	8	4	-4
EOEEOEEO	96	8	12	-4
OEEEOEOE	48	8	6	-4
EEEOEOEO	160	8	20	-4
EOEEOEEO	80	8	10	-4
OEEEOEOE	40	8	5	-4
EEEOEOEO	128	8	16	-4
EOEEEOEO	64	8	8	-4

The q_i values are now uniformly -4 and the eight a_i values $4, 12, 6, 20, 10, 5, 16, 8$ are all distinct, giving a valid 8-cycle.

Note that in this example the cycle steps are driven by the parity sequence directly, not by the residue of the current element modulo h . When $\gcd(g, h) > 1$, the parity of a_i modulo h is not well-defined as a basis for deciding the next step, so the letter-driven interpretation (where w prescribes which steps are O -steps and which are E -steps) is the only coherent one.

(For comparison, the same word at $(g, h) = (3, 2)$ has $d = 5$, $k = 23$, $\gcd(k, |d|) = 1$, giving $q_w = 5$, $a_w = 23$ — the standard Collatz cycle $[23, 74, 37, 116, 58, 29, 92, 46]$.)

Theorem 6.7 (Reduced cycle element identity). *Let w be a parity sequence with o odd steps, e even steps, $d = h^e - g^o$, and $k = k_w(g, h)$. Define*

$$\delta := \begin{cases} \gcd(k, |d|) & \text{if } \gcd(g, h) = 1, \\ \gcd(k_0, k_1, \dots, k_{n-1}, |d|) & \text{otherwise,} \end{cases}$$

where $k_i = k_{w_i}(g, h)$ denotes the structural value of the i -th rotation of w . Then the reduced solution to (6) (with $\gcd(a_w, |q_w|) = 1$) is

$$a_w = \frac{k}{\delta}, \quad q_w = \frac{d}{\delta},$$

Here $a_w > 0$ and q_w has the sign of d . Every other solution with $a > 0$ is a positive-integer multiple of (a_w, q_w) .

Proof. The natural solution (k, d) satisfies (6) by inspection. Dividing both sides by δ gives the reduced solution $(k/\delta, d/\delta)$ with $\gcd(k/\delta, |d/\delta|) = 1$. Any solution (a, q) with $\gcd(a, |q|) = s > 1$ yields a smaller solution $(a/s, q/s)$, so the reduced solution is the unique one with $\gcd(a, |q|) = 1$ and $a > 0$. When $\gcd(g, h) = 1$, Lemma 6.3 guarantees that $\delta = \gcd(k, |d|)$ divides every rotation value, so the single-rotation formula is valid. When $\gcd(g, h) > 1$, the joint gcd is required by Lemma 6.5. $\square$

Theorem 6.8 (Cycle determines q). *Let (g, h) be an integer pair with $\gcd(g, h) = 1$, and let w be a parity sequence with o odd steps, e even steps, $d = h^e - g^o$, and $k = k_w(g, h)$. Given w and (g, h) , the additive term q is not a free parameter: the reduced value*

$$q_w = \frac{d}{\gcd(k, |d|)}$$

is the unique reduced q for which the integer fixed point $a_w = k/\gcd(k, |d|) > 0$ exists. In particular:

1. q_w has the sign of $d = h^e - g^o$.
2. $q_w = 1$ if and only if $|d| \mid k_w(g, h)$.
3. Every other solution with $a > 0$ uses $q = mq_w$ for some $m \in \mathbb{Z}_{>0}$, with corresponding cycle element ma_w .

Proof. By Theorem 6.7, every solution (a, q) with $a > 0$ to $d \cdot a = q \cdot k$ is a positive-integer multiple of the reduced solution (a_w, q_w) . Hence q is constrained to $\{mq_w : m \in \mathbb{Z}_{>0}\}$; the three stated properties follow directly from the definition of $q_w = d/\gcd(k, |d|)$. $\square$

Proposition 6.9 (Polynomial vs. pointwise divisibility). *Polynomial non-divisibility of $h^e - g^o$ into $k_w(g, h)$ in $\mathbb{Z}[g, h]$ does not imply pointwise non-divisibility at any particular integer pair (g, h) . Consequently, no argument based solely on polynomial identities can rule out the existence of integer cycles at a specific basis.*

Proof. The $5x + 1$ cycle at $a = 17$ with word $w = \text{OEE}$ is a concrete witness (Example 10.1): the polynomial $h^7 - g^3$ does not divide $k_w(g, h) = g^2 + gh + h^4$ in $\mathbb{Z}[g, h]$, yet at $(g, h) = (5, 2)$ one has $2^7 - 5^3 = 3$ and $k_w(5, 2) = 51 = 3 \times 17$, so $3 \mid 51$. The cycle exists because of the arithmetic coincidence $2^7 - 5^3 = 3$, which is invisible at the polynomial level. Any argument ruling out Collatz cycles must therefore work at the level of integer evaluations. $\square$

Proposition 6.10 (Successor on k_w around a cycle). *Let (g, h, q) be a basis with $\gcd(g, h) = 1$, and let $w = w_0 w_1 \cdots w_{n-1}$ be a cyclic parity sequence with o odd steps and e even steps in total, and let $w' = w_1 \cdots w_{n-1} w_0$ be the rotation by one step. Then, working in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$,*

$$k_{w'}(g, h) \equiv \begin{cases} g \cdot k_w(g, h) & \text{if } k_w(1, 0) = 1 \text{ (} w \text{ begins with an O-step),} \\ h^{-1} \cdot k_w(g, h) & \text{if } k_w(1, 0) = 0 \text{ (} w \text{ begins with an E-step).} \end{cases}$$

In the O-step case, the unreduced product $g \cdot k_w(g, h)$ promotes the leading monomial $g^{o-1} \mapsto g^o$, and the natural recurrence

$$k_{w'}(g, h) = g \cdot k_w(g, h) + (h^e - g^o)$$

replaces this g^o by h^e — exactly the substitution $g^o \equiv h^e \pmod{h^e - g^o}$ — keeping the degree at $o - 1$ in g . In the E-step case, h^{-1} exists in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$ because $\gcd(g, h) = 1$ implies $\gcd(h, h^e - g^o) = 1$. The modulus $h^e - g^o$ is the cycle denominator from Theorem 6.7, so this operation is only natural in the context of a fixed cycle (o, e) . As a consequence, if w is primitive (not a proper power of a shorter word), the n rotations of w produce at most n distinct values of $k_{w'}(g, h)$ in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$. For a non-primitive w of period $p \mid n$, there are at most p distinct values.

Corollary 6.11 (Canonical cycle basis). *Let (g, h) be an integer pair with $\gcd(g, h) = 1$, let w be a parity sequence with o odd and e even steps, and let $d = h^e - g^o$, $k = k_w(g, h)$, $\delta = \gcd(k, |d|) > 0$. The canonical pair*

$$q_w := \frac{d}{\delta}, \quad a_w := \frac{k}{\delta}$$

is the unique reduced solution to $d \cdot a = q \cdot k$ with $a_w > 0$ (i.e. $\gcd(a_w, q_w) = 1$). Here $a_w > 0$ and q_w has the sign of d . Every solution with $a > 0$ is a positive-integer multiple of (a_w, q_w) .

Proof. This is the content of Theorem 6.7. $\square$

Remark 6.12. When $d \mid k_w$, the canonical basis has $q_w = 1$, recovering the standard Collatz setting. A worked example ($w = OEEOEOEE$, $q_w = 5$, $a_w = 29$) appears in Section 8.

7 Cycle Flavours

The cycle element identity $d \cdot a = q \cdot k$ has infinitely many integer solutions, related by overall scaling and sign. Different conventions on the signs and integrality of a and q give rise to what we call *cycle flavours*. These are not distinct cycles: they are the same underlying algebraic object viewed through different choices of normalisation.

Flavour	a	q	$\gcd(a , q)$
Natural	$k > 0$	d (any sign)	≥ 1
Reduced	$k / \gcd(k, d) > 0$	$d / \gcd(k, d)$ (any sign)	1
Negative	< 0	> 0	1
Rational	k/d (any sign)	1	—

The equivalences between flavours all follow from the symmetry of $d \cdot a = q \cdot k$:

- **Natural $\leftrightarrow$ scaled:** multiplying both a and q by any $m \in \mathbb{Z}_{>0}$ gives another valid solution. The natural solution (k, d) is the representative with $a = k$.
- **Reduced and Negative:** both have $\gcd(|a|, |q|) = 1$ but differ in sign convention. A reduced cycle has $a > 0$ with q inheriting the sign of d ; a negative cycle has $a < 0$ and $q > 0$. The same underlying equation $d \cdot a = q \cdot k$ is satisfied in both cases: negating both a and q simultaneously is always valid, so every reduced cycle with $q < 0$ corresponds to a negative cycle with $q > 0$, and every reduced cycle with $q > 0$ has no negative counterpart with $\gcd = 1$ (since negating gives $a < 0$, $q < 0$, which violates $q > 0$).
- **Rational:** setting $q = 1$ gives $a = k/d \in \mathbb{Q}$. This is an integer exactly when $d \mid k$, i.e. when the reduced cycle has $q_w = 1$. The sign of a matches the sign of d .

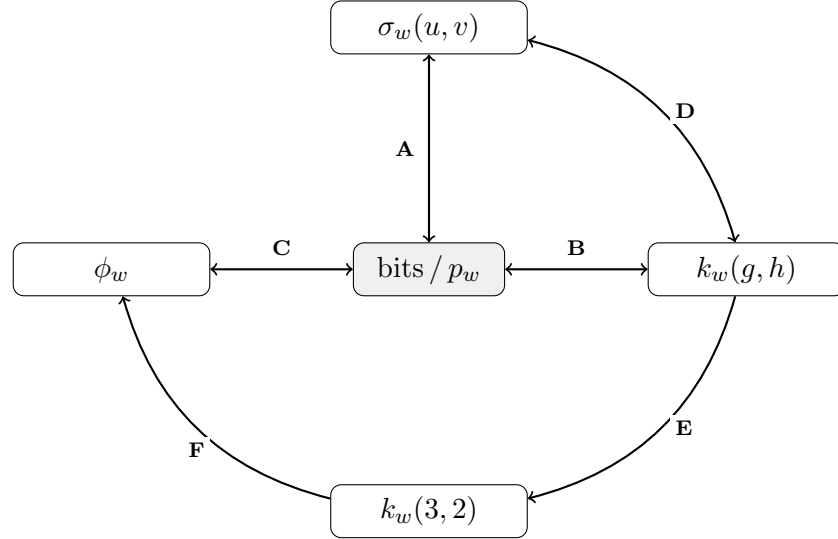
The preferred flavour in this paper is *reduced* with $a > 0$. The mythical Collatz cycles are the reduced cycles at $(g, h, q) = (3, 2, 1)$ with $o > 1$: rational cycles at $q = 1$ that are also integers, with $a > 0$.

8 Representations of a Parity Sequence

Definition 8.1 (Orbit parity sequence ϕ_w). Let w be a parity sequence with o odd steps, e even steps, and $k_w(3, 2) = k$. Set $q = 2^e - 3^o$ and define the sequence $a_0 = k$, $a_{i+1} = \Phi_{\text{step}}(a_i)$ under the $(3, 2, q)$ Collatz map. The *orbit parity sequence* of w is the parity sequence

$$\phi_w := (s_0, s_1, \dots, s_{n-1}), \quad s_i = \begin{cases} O & \text{if } a_i \text{ is odd,} \\ E & \text{if } a_i \text{ is even.} \end{cases}$$

Every parity sequence w of length n with o odd steps lives simultaneously in five equivalent representations. The canonical integer encoding (bits, p_w) sits at the centre; four structural representations — $\sigma_w(u, v)$, $k_w(g, h)$, $k_w(3, 2)$, and ϕ_w (Definition 8.1) — surround it. The diagram below serves as a navigational frame for the subsections that follow; each subsection develops one or more nodes and the edges connecting them.



Edge	Pair	Forward	Inverse
A	$\text{bits} \leftrightarrow \sigma_w$	$\sigma_w(u, v) = \sum_{j=0}^{o-1} u^{o-1-j} v^{p_j}$	$p_w = \sigma_w(1, 2) + 2^n$
B	$\text{bits} \leftrightarrow k_w(g, h)$	$k_w(g, h) = \sum_{j=0}^{o-1} g^{o-1-j} h^{p_j}$	$p_w = 2^{o-1} k_w(\frac{1}{2}, 2) + 2^n$
C	$\text{bits} \leftrightarrow \phi_w$	$\phi_w[i] = O \text{ if } b_i = 1, \text{ else } \phi_w[i] = E$	$b_i = 1 \text{ if } \phi_w[i] = O, \text{ else } b_i = 0$
D	$\sigma_w \leftrightarrow k_w(g, h)$	$k_w(g, h) = \sigma_w(gh, h)/h^{o-1}$	$\sigma_w(u, v) = k_w(u/v, v) \cdot v^{o-1}$
E	$k_w(g, h) \rightarrow k_w(3, 2)$	$(g, h) \mapsto (3, 2)$	(via F then C then B)
F	$k_w(3, 2) \rightarrow \phi_w$ (Def. 8.1)	$q = 2^e - 3^o$, $a_0 = k_w(3, 2)$; iterate $(3, 2, q)$ map	(via C then B)

8.1 The p -value (edges B, C; centre node)

Since k_w is independent of trailing E -steps (Remark 2.8), it does not uniquely identify a parity sequence. A simple specialisation recovers a canonical integer encoding, and a sentinel bit resolves the trailing- E ambiguity.

Proposition 8.2 (Bit-encoding of parity sequences). *Let $w = b_0b_1 \cdots b_{n-1}$ be a parity sequence of length n , where $b_i = 1$ if the i -th step is an O -step and $b_i = 0$ if it is an E -step. Then*

$$N_w := 2^{o-1} \cdot k_w\left(\frac{1}{2}, 2\right) = \sum_{i=0}^{n-1} b_i 2^i,$$

the non-negative integer whose binary representation encodes w directly (LSB-first, $O = 1$, $E = 0$).

Proof. The j -th term of $k_w(g, h)$ is $g^{o-1-j} h^{\kappa_j}$. Evaluating at $g = 1/2$, $h = 2$ and multiplying by 2^{o-1} :

$$2^{o-1} \cdot \left(\frac{1}{2}\right)^{o-1-j} \cdot 2^{\kappa_j} = 2^{j+\kappa_j}.$$

Since κ_j counts E -steps before the j -th O -step and j counts O -steps before it, $j + \kappa_j = p_j$ is the position of the j -th O -step, i.e. the index i with $b_i = 1$. Summing over j gives $\sum_j 2^{p_j} = \sum_i b_i 2^i = N_w$. $\square$

Sequences differing only in trailing E -steps (i.e. trailing $b_i = 0$) share the same N_w . To resolve this ambiguity we append a sentinel bit at position n .

Definition 8.3 (p -value). The p -value of a parity sequence $w = b_0b_1 \cdots b_{n-1}$ is

$$p_w := \sum_{i=0}^{n-1} b_i 2^i + 2^n = 2^{o-1} \cdot k_w\left(\frac{1}{2}, 2\right) + 2^n.$$

Theorem 8.4 (p -value uniqueness). *The map $w \mapsto p_w$ is a bijection from the set of all finite parity sequences over $\{O, E\}$ (including the empty sequence) to the set of integers ≥ 1 . The empty sequence has $p_\varepsilon = 2^0 = 1$. For non-empty w , w is uniquely recovered from p_w by writing p_w in binary: the highest set bit (at position n) records the length, and bits $0, \dots, n-1$ give the sequence with $1 = O$ and $0 = E$.*

Proof. The empty sequence has $n = 0$, $N_\varepsilon = 0$, and $p_\varepsilon = 0 + 2^0 = 1$. For non-empty w , given $p_w \geq 2$, let $n = \lfloor \log_2 p_w \rfloor \geq 1$ (the position of the highest set bit). Then bits $0, \dots, n-1$ of p_w coincide with N_w and uniquely determine the positions of all O -steps, hence the full sequence w . The map is therefore injective. Surjectivity onto integers ≥ 1 follows since every positive integer has a unique highest set bit at some position $n \geq 0$: $n = 0$ gives the empty sequence, and $n \geq 1$ gives the parity sequence whose i -th step is O iff bit i of p_w is 1. $\square$

Remark 8.5 (Successor as cyclic bit rotation). The successor $w \mapsto w'$ (Proposition 6.10) acts on the p -value as a *right* rotation of the lower n bits. Reading the data bits in the conventional MSB-first order as $b_{n-1}b_{n-2} \cdots b_1b_0$, the right rotation moves the rightmost bit b_0 to the leftmost data position:

$$p_{w'} = 2^n + b_0 2^{n-1} + b_{n-1} 2^{n-2} + \cdots + b_1.$$

The sentinel bit at position n is unchanged, reflecting the fact that all cycle representatives have the same length n .

8.2 The σ -polynomial (edges A, D)

Change of variables

The σ -polynomial arises from $k_w(g, h)$ by the substitution $g = u/v$, $h = v$, followed by clearing the denominator v^{o-1} : $\sigma_w(u, v) = k_w(u/v, v) \cdot v^{o-1} \in \mathbb{Z}[u, v]$. The inverse substitution $u = gh$, $v = h$ recovers $k_w(g, h) = \sigma_w(gh, h)/h^{o-1}$.

Definition 8.6 (σ -polynomial). For a parity sequence w of length n with o odd steps and O -step positions $p_0 < \dots < p_{o-1}$, define

$$\sigma_w(u, v) := k_w\left(\frac{u}{v}, v\right) \cdot v^{o-1} = \sum_{j=0}^{o-1} u^{o-1-j} v^{p_j} \in \mathbb{Z}[u, v].$$

The exponent of u in the j -th term counts the O -steps *after* position j ; the exponent of v is the bit-position of the j -th O -step. Specialising to $u = 1$, $v = 2$ recovers the bit-encoding: $N_w = \sigma_w(1, 2)$, and consequently $p_w = \sigma_w(1, 2) + 2^n$ (edge A inverse).

The successor operation on σ

Each step of the Collatz map acts on $\sigma_w(u, v)$ by an elegant polynomial operation [1]. Define

$$\gamma_w := \begin{cases} 1 & \text{if } p_0 = 0 \text{ (i.e. } w \text{ begins with an } O\text{-step),} \\ 0 & \text{otherwise.} \end{cases} \quad (\gamma)$$

Equivalently, $\gamma_w = 1$ if and only if the leading term $u^{o-1}v^{p_0}$ of σ_w has v -exponent zero. Define

$$\text{succ}_\sigma(\sigma_w(u, v)) := u^{\gamma_w} \cdot \sigma_w(u, v) \cdot v^{-1}.$$

Then $\text{succ}_\sigma(\sigma_w(u, v)) = \sigma_{w'}(u, v)$, where w' is the parity sequence obtained by rotating w left by one step (moving the first bit to the last position).

Remark 8.7. Adopting the convention $0^0 = 1$, we have $\gamma_w = \sigma_w(1, 0)$: each term $1^{o-1-j} \cdot 0^{p_j}$ vanishes unless $p_j = 0$, which occurs precisely when $j = 0$ and w begins with an O -step.

The successor formula is valid in any ring where $u^o = 1$ and $v^n = 1$: the cycle closes after n steps since $v^{-n} = 1$, returning to σ_w . Complex roots of unity are one natural realisation of these constraints, but the algebraic structure depends only on these two relations.

Evaluation at roots of unity

Proposition 8.8. Let $\omega_o = e^{2\pi i/o}$ and $\omega_n = e^{2\pi i/n}$ be primitive o -th and n -th roots of unity respectively. The evaluation

$$\sigma_w(\omega_o, \omega_n) = \sum_{j=0}^{o-1} \omega_o^{o-1-j} \omega_n^{p_j}$$

expresses the parity sequence as a linear combination of n -th roots of unity, weighted by powers of ω_o . Each step of the cycle acts as

$$\text{succ}_\sigma(\sigma_{w_k}(\omega_o, \omega_n)) = \omega_o^{\gamma_{w_k}} \cdot \sigma_{w_k}(\omega_o, \omega_n) \cdot \omega_n^{-1},$$

a multiplication by $\omega_o^{\gamma_{w_k}}$ (recording whether step k is an O -step) composed with a rotation ω_n^{-1} of the phase. Since $\omega_o^o = 1$ and $\omega_n^n = 1$, the cycle closes after exactly n steps. See [1] for the full development.

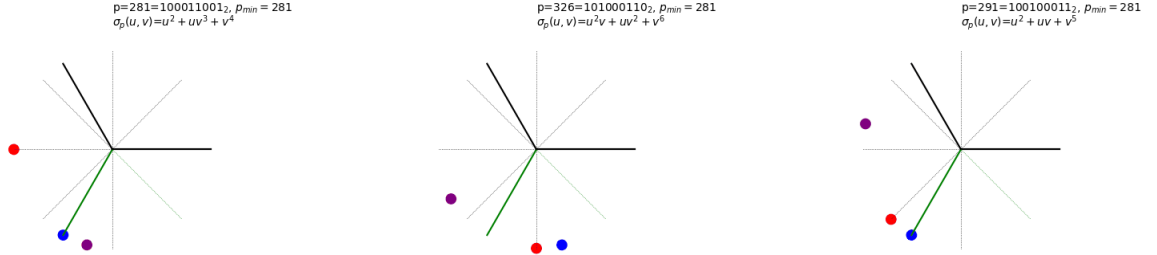


Figure 1: Three successive steps of the σ -successor for $w = OEEOOEEE$ ($n = 8$, $o = 3$), evaluated at (ω_3, ω_8) . Coloured dots show the three monomials of σ_w ; the green arrow is their sum (the value of σ_w as a complex number). *Left* (step 0, O -step, $\gamma = 1$): $\sigma = u^2 + uv^3 + v^4$. The O -step multiplies by ω_3 in addition to rotating by ω_8^{-1} , changing the u -weights as well as the phase. *Centre* (after two E -steps): $\sigma = u^2v + uv^2 + v^6$ — the dots have rotated twice; the sum (green arrow) has changed direction. *Right* (after the second O -step, $\gamma = 1$): $\sigma = u^2 + uv + v^5$. The O -step multiplies by ω_3 in addition to rotating by ω_8^{-1} , redistributing the u -weights and visibly changing the triangle of dots.

8.3 Worked example (all edges, $w = OEEOEOEE$)

The remaining nodes $k_w(3, 2)$ and ϕ_w are reached via edges E and F respectively; the return path from ϕ_w runs via **C** then **B** then **E**.

$$\mathbf{C}^{-1} \quad b_8 b_7 \cdots b_0 = 1, 0, 0, 1, 0, 1, 0, 0, 1 \quad (O=1, E=0, b_8=1 \text{ sentinel})$$

$$O\text{-positions } p_0=0, p_1=3, p_2=5$$

$$\mathbf{A} \quad \sigma_w(u, v) = u^2 + u v^3 + v^5$$

$$\mathbf{A}^{-1} \quad p_w = \sigma_w(1, 2) + 2^8 = 41 + 256 = 297 = 100101001_2$$

$$\mathbf{D} \quad k_w(g, h) = \sigma_w(gh, h)/h^2 = g^2 + gh^2 + h^3$$

$$\mathbf{B}^{-1} \quad p_w = 2^{o-1} k_w\left(\frac{1}{2}, 2\right) + 2^8 = 41 + 256 = 297 = 100101001_2 \quad \checkmark$$

$$\mathbf{E} \quad k_w(3, 2) = 3^2 + 3 \cdot 2^2 + 2^3 = 29$$

$$\mathbf{F} \quad q = 5, a_0 = 29; \text{ iterate } (3, 2, 5):$$

$$29 \xrightarrow{O} 92 \xrightarrow{E} 46 \xrightarrow{E} 23 \xrightarrow{O} 74 \xrightarrow{E} 37 \xrightarrow{O} 116 \xrightarrow{E} 58 \xrightarrow{E} 29 \Rightarrow \phi_w = OEEOEOEE$$

$$\mathbf{C} \quad b_8 b_7 \cdots b_0 = 1, 0, 0, 1, 0, 1, 0, 0, 1 \quad (b_8=1 \text{ sentinel encodes } n=8)$$

$$\mathbf{B} \quad \kappa_j = (0, 2, 3), \quad k_w(g, h) = g^2 + gh^2 + h^3 \quad \checkmark$$

Since $\gcd(5, 29) = 1$, the canonical basis gives $q_w = 5$, $a_w = 29$: no $3x+1$ cycle, but an integer $3x+5$ cycle at $a = 29$.

9 What the Abstract Cycle Tells Us — and What It Does Not

The preceding sections have established that a parity sequence w admits many equivalent representations: as the orbit parity sequence ϕ_w , as a binary integer p_w , as the σ -polynomial $\sigma_w(u, v)$, as the structural polynomial $k_w(g, h)$ over an arbitrary basis, and as the integer $k_w(3, 2)$. When w is a cycle word, it carries additionally the cycle flavour table of Section 7. Each of these representations illuminates a different facet of the same object.

Ironically, one of the least revealing representations of a cycle is the one most commonly encountered in the literature: a cycle expressed as a list of integers in a specific basis (g, h, q) . When $k = k_w(g, h)$ and $d = h^e - g^o$ are collapsed into a single ratio $a = qk/d$, most of the algebraic structure disappears. The cycle element a is just a number. The richness encoded in k_w as a polynomial in g and h — its monomial structure, the κ -tuple, the geometric sum form for Steiner words — is invisible. That the structure survives at all, and that a is an integer, is one of the small mysteries of mathematics; but the *why* is not apparent from the integer alone.

By contrast, the abstract representations make structure explicit. The κ -tuple encodes exactly how E-steps are distributed before each O-step. The structural polynomial $k_w(g, h)$ makes the dependence on the basis parameters transparent: one can see immediately how the cycle element scales as g or h varies, and how cycles at different bases relate to one another. The σ -polynomial connects the combinatorial structure of w to geometry: the monomials are points in a lattice, and the successor operation is a rotation. The p -value gives a canonical integer encoding that is independent of any basis.

All integer Collatz-like cycles are mere encodings of an abstract cycle

Once a parity sequence w and an encoding basis (g, h) are chosen, everything else is determined. The cycle element $a_w = k_w(g, h) / \gcd(k_w(g, h), |d|)$ and the additive parameter $q_w = d / \gcd(k_w(g, h), |d|)$ follow with no further freedom. There are no free integer cycles: every integer cycle in any basis is an encoding of some abstract parity sequence w , with the parameters (a, q) fixed by (w, g, h) through the reduced cycle element identity. The space of abstract parity sequences is therefore in bijection with the space of reduced integer cycles across all bases.

This perspective also reveals shared structure across different bases. Two bases (g_1, h_1, q_1) and (g_2, h_2, q_2) that yield the same parity sequence w as a cycle are encoding the same abstract object. The structural polynomial $k_w(g, h)$ makes this commonality explicit: any algebraic identity that holds in $\mathbb{Z}[g, h]$ — such as the factorisation $k_w(g, h) = h^e - g^o$ on the variety $g = h^2 - 1$ — is inherited by every basis that specialises to that variety.

The polynomial form can characterise individual bases

That said, the polynomial form can establish non-trivial facts that are *specific to the* $(3, 2, 1)$ *encoding*. A striking example is the $2a+1$ composition identity: for the Steiner words $\text{St}(\alpha, 1)$

and $\text{St}(1, \beta)$, the identity

$$\Phi_{\text{St}(1, \beta)}(\Phi_{\text{St}(\alpha, 1)}(a)) = \Phi_{\text{St}(\alpha+1, \beta+2)}(2a+1)$$

holds for all $\alpha \geq 1$, $\beta \geq 0$, and all admissible a . In [4] we prove — working entirely in $\mathbb{Z}[g, h]$ via the structural polynomial — that among all bases (g, h, q) with $g, h \geq 2$ and $g \neq h$, this identity characterises $(g, h, q) = (3, 2, 1)$ uniquely. The proof forces $h = 2$ from the coefficient of a , then $g = 3$ by requiring the constant-term equation to hold for all α , and finally $q = 1$ by substitution. This is a purely polynomial argument: no numerical verification, no modular arithmetic, no appeal to orbit dynamics. The structural polynomial framework is the natural setting for such uniqueness results.

The abstract framework has limits, however. It does not, by itself, answer the question of whether the standard Collatz map $(3, 2, 1)$ has a non-trivial cycle. That question requires understanding the modular arithmetic of $3x+1$, $x/2$ — the residue structure, the admissibility constraints, the interplay between the parity of $3^o a + k_w(3, 2)$ and the required E-steps.

The limits are not total, however. The structural polynomial does carry modular information, and only a finite truncation is needed to extract it. Every monomial $g^{o-1-j} h^{\kappa_j}$ with $j \geq 3$ has $\kappa_j \geq 3$ and therefore contributes a multiple of h^3 to $k_w(g, h)$. The first three terms alone,

$$g^{o-1} + g^{o-2} h^{\kappa_1} + g^{o-3} h^{\kappa_2} \pmod{h^3},$$

determine $k_w(g, h) \pmod{h^3}$ completely — for any basis. At $(g, h) = (3, 2)$ this reads off the mod-8 character of the cycle element; at $(g, h) = (8, 3)$ the same truncation reads off the mod-27 character. The polynomial carries both facts simultaneously, as specialisations of a single identity in $\mathbb{Z}[g, h]$.

What the polynomial cannot tell us is whether the mod-8 structure of the $(3, 2, 1)$ map, or the mod-27 structure of the $(8, 3, 1)$ map, is the right tool for attacking the cycle question in each case. Those residue structures are quite different — the arithmetic accidents that make one basis tractable need not have any analogue in the other — and the details almost certainly matter. The abstract framework reveals the *form* of the modular constraint; deciding its *utility* requires engaging with the specific arithmetic of the basis in question.

In short: the abstract cycle framework is the right language for understanding *what* cycles are and *how* they relate to one another across bases. It is not sufficient on its own for understanding *whether* a particular cycle exists for a *specified* q : for any parity word w and any (g, h) , a cycle always exists if one is prepared to let q be determined by (w, g, h) via the reduced cycle element identity, but whether that q equals a pre-specified value — such as $q = 1$ for the standard Collatz map — is a question the abstract form alone cannot settle. For that, one must engage with the concrete arithmetic of the map. The abstract framework reaches further into that arithmetic than it might first appear, but both levels of description are necessary, and they overlap more than the literature typically acknowledges.

10 Examples

Example 10.1 (The $5x + 1$ cycle at 17). Under the Syracuse $5x + 1$ map (basis $g = 5$, $h = 2$, $q = 1$), the odd integer 17 is a periodic point. Its orbit is

$$17 \xrightarrow{O} 86 \xrightarrow{E} 43 \xrightarrow{O} 216 \xrightarrow{E^3} 27 \xrightarrow{O} 136 \xrightarrow{E^3} 17.$$

The parity sequence is $w = OEEOEEEOEEE$, with $o = 3$, $e = 7$.

The κ -tuple is $\kappa_0 = 0$, $\kappa_1 = 1$, $\kappa_2 = 4$ (the three O -steps are preceded by 0, 1, and 4 even steps respectively). Note that $(\kappa_0, \kappa_1, \kappa_2) = (0, 1, 4) \neq (0, 1, 2)$, so w is *not* a Steiner word and the geometric polynomial theorem does not apply directly. The structural polynomial is

$$k_w(g, h) = g^2h^0 + g^1h^1 + g^0h^4 = g^2 + gh + h^4.$$

Evaluating at $(g, h) = (5, 2)$:

$$k_w(5, 2) = 25 + 10 + 16 = 51, \quad h^e - g^o = 2^7 - 5^3 = 128 - 125 = 3.$$

By Theorem 6.7,

$$a = \frac{1 \cdot 51}{3} = 17. \quad \checkmark$$

The near-miss $h^e - g^o = 3$ is what makes this cycle arithmetically possible: $2^7 - 5^3 = 3$ is a specific arithmetic accident with no analogue for the Collatz basis $(g, h) = (3, 2)$. This also illustrates Proposition 6.9: the polynomial $h^7 - g^3$ does not divide $g^2 + gh + h^4$ in $\mathbb{Z}[g, h]$, yet divisibility holds at $(5, 2)$ because of this accident.

Example 10.2 (The $g = h^2 - 1$ family). When divisibility *does* hold as a polynomial identity, it explains cycles across an entire family of bases at once. Consider $\text{St}(1, 1)^3$ (three copies of OEE), with $o = 3$, $e = 6$. By the composition rule:

$$\begin{aligned} k_{\text{St}(1,1)^2}(g, h) &= g + h^2, \\ k_{\text{St}(1,1)^3}(g, h) &= g^2 + gh^2 + h^4. \end{aligned}$$

On the locus $g = h^2 - 1$ one checks that $(g^2 + gh^2 + h^4) - (h^6 - g^3) = (g - h^2 + 1)(g^2 + gh^2 + h^4)$ vanishes identically on the locus $g = h^2 - 1$ (since $g - h^2 + 1 = 0$ there), so $k_{\text{St}(1,1)^3} = h^6 - g^3$ and the cycle element $a = q \cdot k / (h^6 - g^3) = q$ for every such basis. At $q = 1$:

- $h = 2$, $g = 3$: orbit $1 \xrightarrow{O} 4 \xrightarrow{EE} 1$, three times.
- $h = 3$, $g = 8$: orbit $1 \xrightarrow{O} 9 \xrightarrow{EE} 1$, three times.

Example 10.3 (Non-Collatz parity sequence). This example relaxes the requirement that the κ -tuple be strictly increasing. In the integer domain this corresponds to relaxing the rule that $gx + q$ is applied only to odd integers; in both settings the result is an “illegal” cycle that does not arise in standard Collatz dynamics.

Consider the parity sequence $w = OEEOOEEEE$, with $o = 3$, $e = 5$. The κ -tuple is $\kappa_0 = 0$, $\kappa_1 = 2$, $\kappa_2 = 2$: both the second and third O -steps are preceded by exactly 2 even steps. The structural polynomial is

$$k_{OEEOOEEEE}(g, h) = g^2h^0 + g^1h^2 + g^0h^2 = g^2 + h^2(g + 1).$$

This w is *not* a valid standard Collatz parity sequence. The equal values $\kappa_1 = \kappa_2 = 2$ mean that two consecutive O -steps occur without any intervening E -step—the subsequence OO appears. Under standard Collatz, an O -step must be applied to an odd integer, so two consecutive O -steps are impossible: after $3n + 1$ the result is even and must be halved before another O -step can occur. Equivalently, valid Collatz sequences require $\kappa_0 < \kappa_1 < \cdots < \kappa_{o-1}$ (strictly increasing).

If one relaxes this single condition—allowing $\kappa_j \leq \kappa_{j+1}$ —and correspondingly drives the $3x + 1$ and $x/2$ operations by the letters of w rather than the parity of the current element, then cycles can exist. At $(g, h) = (3, 2)$ we get $k_w(3, 2) = 3^2 + 2^2(3 + 1) = 9 + 16 = 25$ and $h^e - g^o = 2^5 - 3^3 = 5$, so Theorem 6.7 yields cycle element $a = 25/5 = 5$. The orbit from $a = 5$ is

$$5 \xrightarrow{O} 16 \xrightarrow{E} 8 \xrightarrow{E} 4 \xrightarrow{O} 13 \xrightarrow{O} 40 \xrightarrow{E} 20 \xrightarrow{E} 10 \xrightarrow{E} 5,$$

which does return to 5. The single illegal step is $4 \xrightarrow{O} 13$: applying $3x + 1$ to the even integer 4, forced by the repeated value $\kappa_1 = \kappa_2 = 2$.

Remark 10.4. The sequence $OEEOOEEE$ above has p -value $N_w + 2^8 = 25 + 256 = 281$: the integer 281 encodes the sequence directly in binary (with sentinel bit at position 8), and the glitched orbit was constructed by reading the bits of $281 - 256 = 25$ as a parity sequence.

11 Related Work

The structural polynomial $k_w(g, h)$ is closely related to the k -parameter appearing in the standard affine representation of Collatz orbits, where it appears as an entry in the affine transformation matrix. The Steiner word framework was developed in [2] as a natural decomposition of Syracuse orbits; the present paper provides its polynomial algebraic foundation.

The σ -polynomial $\sigma_w(u, v)$ and its successor operation at complex roots of unity were introduced in [1]; the present paper situates them within the broader five-representation framework and makes their connection to $k_w(g, h)$ via the change of variables $g = u/v$, $h = v$ explicit.

The composition rule (Theorem 4.1) and cycle element identity (Theorem 6.7) are applied in the companion paper [4] to prove a uniqueness theorem for the $2a + 1$ Steiner composition identity and to characterise its domain constraint in terms of 2-adic valuations. Structural constraints on Collatz cycles derived from Steiner sentence analysis appear in [3].

References

- [1] J. Seymour, *Exploring Cyclic Structures in Polynomial Rings with Applications to the Collatz Conjecture*, preprint, 2025. <https://wildducktheories.github.io/collatz/papers/14-uv-polynomial/uv-polynomial.pdf>
- [2] J. Seymour, *Steiner Word Decomposition and Branch Length Distribution*, preprint, 2026. <https://wildducktheories.github.io/collatz/papers/67-branch-length-distribution/67-branch-length-distribution.pdf>

- [3] J. Seymour, *Structural Constraints on Collatz Cycles via Steiner Sentence Analysis*, unpublished, 2026.
- [4] J. Seymour, *The $2a + 1$ Composition Identity: Uniqueness and Domain Constraint*, unpublished, 2026.